DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CONNECTED MOBILITY, LOGISTICS & LOT SECURITY

Connected Vehicle Fleet Monitoring and Predictive Maintenance System

From reactive repairs to data-driven maintenance

PROJECT REVIEW
Domain: CONNECTED MOBILITY, LOGISTICS & LOT SECURITY
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Introduction

Connected vehicles stream sensor and location data continuously. Fleets can use that data to fix problems **before** they cause breakdowns.

Connected fleets

IoT devices and telematics report speed, engine temperature, battery, fuel, tires, vibration and GPS.

The challenge

Managers lack a single view of vehicle health, so failures are found late and downtime is unexpected.

The approach

Combine live monitoring with machine learning to move from reactive to predictive maintenance.

Abstract

An intelligent fleet-management platform that continuously monitors connected vehicles, analyzes operational and sensor data, detects abnormal behavior and predicts maintenance needs before major failures occur. Separate dashboards serve fleet managers, maintenance teams and administrators.

1 Collect

IoT devices, simulated telematics streams or historical datasets

2 Analyze

Health score, fuel efficiency, utilization, driving patterns, fault frequency

3 Predict

Random Forest, XGBoost, Logistic Regression, LSTM, Isolation Forest

4 Act

Threshold and ML-based alerts with recommended service actions

Parameters monitored: speed, engine RPM and temperature, battery voltage, fuel level, oil and tire pressure, brake temperature, vibration, mileage, GPS location, diagnostic trouble codes

Literature Review Table

Paper	Method	Dataset	Result	Limitation	Gap Identified
Arena et al. (2022) [1]	Systematic review of statistical, stochastic and AI methods for automotive predictive maintenance	Published studies	Deep learning is more accurate but needs more data than traditional ML	Real datasets are scarce; methods are hard to compare	No fleet-level system joining monitoring, prediction and maintenance
Killeen et al. (2019) [2]	IoT architecture with vehicle, server-leader and root nodes for fleet predictive maintenance	Public transport buses (STO, Gatineau, Canada)	Architecture and prototype for fleet-wide and on-board analytics	One bus fleet; prototype stage	Little on maintenance scheduling and role-based dashboards

Literature Review Table (continued)

Paper	Method	Dataset	Result	Limitation	Gap Identified
Chaudhuri (2018) [3]	Hierarchical modified fuzzy SVM, compared with logistic regression, Random Forest and SVM	Garage repair records of a vehicle fleet	About 96.6% accuracy vs 66.9% (LR), 70.9% (RF), 80.7% (SVM)	Repair history, not live sensor streams; class imbalance	No real-time sensor monitoring or component-level alerts
Shafi et al. (2018) [4]	SVM, Decision Tree, Random Forest and k-NN for remote vehicle health monitoring	Vehicle subsystem data (ignition, cooling, fuel)	SVM best; accuracy above 96% as summarised in [1]	Subsystem fault classification	Fleet analytics, service records and alerts not integrated
Our Work	MQTT and Kafka streaming; Random Forest, XGBoost, Isolation Forest, LSTM	Synthetic telemetry: 50–200 vehicles, 10,000+ records	Planned evaluation: precision, recall, F1-score	Simulated data in the first phase	Monitoring, prediction, alerts and maintenance in one platform

Literature Summary and Research Gap

Common strengths

- ML predicts failures from sensor and maintenance data
- IoT architectures allow fleet-wide, real-time collection
- SVM and ensemble models report accuracy above 96%
- Predictive approaches go beyond reactive and scheduled servicing

Common limitations

- Real datasets are scarce and hard to compare
- Deep learning needs large data volumes
- Studies rely on garage records or one subsystem
- Fleet systems stay at prototype level

Research gap: each study covers one part, such as architecture, classification or review. This project joins real-time monitoring, ML prediction, alerts and maintenance management in one role-based platform.

Existing System

Current practice

- Manual inspections with paper or spreadsheet service logs
- Reactive repairs after a breakdown
- Preventive servicing at fixed time or mileage intervals
- Basic GPS tracking that shows location only
- Research prototypes limited to garage records or one subsystem

Drawbacks

- Unexpected breakdowns and unplanned downtime
- Parts replaced too early or too late, raising cost
- No live view of engine, battery, tire and brake health
- Problems found by drivers or manual checks, not by data
- No single fleet view of service history and alerts

Proposed System

01

Vehicle Sensors

Speed, RPM, temperature, battery, tires, GPS

02

IoT Gateway

Device publishes readings over MQTT

03

Streaming

Apache Kafka topics buffer the telemetry

04

Ingestion

Spring Boot telemetry service validates data

05

Storage

PostgreSQL plus a time-series database

06

Analytics and ML

Health score, anomaly detection, failure risk (Python)

07

Alerts

Threshold and ML-risk alerts with recommended action

08

Dashboards

Views for fleet managers, maintenance teams and admins

Approach: streaming ingestion feeds analytics and ML, and the results reach users through dashboards and alerts.

Technologies and Algorithms

Frontend

React.js, TypeScript, Chart.js / Recharts, Leaflet maps

Backend and Security

Java 21, Spring Boot, Spring REST, JWT and OAuth2

Data Layer

PostgreSQL, InfluxDB / TimescaleDB, Redis cache

IoT and Messaging

MQTT for devices, Apache Kafka for streaming

ML Algorithms

Random Forest [5], XGBoost [6], Isolation Forest [7], LSTM [8]

DevOps

Docker, Kubernetes, GitHub Actions, Prometheus, Grafana

Scope of the Project

Monitor

Live fleet dashboard
GPS and route tracking
Vehicle health score
Vehicle master data

Predict

Fault and anomaly detection
Predictive maintenance model
Threshold and ML alerts
Component failure risk

Manage

Maintenance scheduling
Service history and cost
Fleet analytics and reports
Role-based access (JWT)

Users: Fleet Manager, Maintenance Engineer, Fleet Operator, Administrator

Implementation Scope

50–200

simulated vehicles

10,000+

sensor records, with deliberate failure patterns

Seconds

to minutes between telemetry readings

Demo: a healthy vehicle gradually develops abnormal sensor patterns, then the system raises an AI-generated maintenance-risk alert with a recommendation.

Java + Spring Boot · React · Python ML · PostgreSQL · Kafka / MQTT

Conclusion and Future Scope

Conclusion

- The system joins IoT, streaming, analytics, ML and dashboards in one platform
- The literature shows ML-based maintenance prediction outperforms reactive and fixed-schedule servicing
- It addresses the gap: monitoring, prediction, alerts and maintenance management together
- Expected outcome: less unexpected downtime and earlier fault detection

Future scope

- LSTM time-series models and remaining useful life
- Real telematics or OBD-II data from live vehicles
- Edge processing on the vehicle
- Driver behavior analytics
- Mobile app for technicians

References

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